

Regression Trees

Daniel Lawson University of Bristol

Lecture 03.1.2 (v4.0.0)

- ▶ How do we use a tree to perform regression?
 - ▶ What is the “natural” loss function?
 - ▶ What pitfalls might exist in this approach?

Regression Trees

- ▶ Decision trees can also be used for regression
- ▶ There is a nice relationship between the loss function of classification, and “mean squared error” in regression

Regression Trees (simple version)

- ▶ Regression trees are constructed identically to classification trees
- ▶ Decision can follow information or squared loss
- ▶ Prediction is the average $\hat{y}_j | \{i \in c_j\} = \bar{y}_{c_j}$ inside the leaf j
- ▶ Comparing to classification: if $y \in \{0, 1\}$ this reduces to:

$$R_j = \frac{1}{N_j} \sum_{i \in c_j} (y_i - \hat{y}(x_i))^2 \quad (1)$$

$$= \frac{1}{N_j} \sum_{i \in c_j} \left(y_i^2 - 2y_i\hat{y}(x_i) + \hat{y}(x_i)^2 \right) \quad (2)$$

- ▶ Which since $\hat{y} = \bar{y}$, simplifies to:

$$R = \bar{y} - 2\bar{y}^2 + \bar{y}^2 \quad (3)$$

$$= \bar{y}(1 - \bar{y}) \quad (4)$$

Comparing Regression Trees to Classification Trees

- ▶ Similarly the Gini index is:

$$G = \sum_{j=1}^J p_j(1 - p_j) \quad (5)$$

$$= \bar{y}(1 - \bar{y}) \quad (6)$$

- ▶ So regression trees and classification trees use equivalent loss functions in CART.

General Regression Trees

- ▶ Within each decision node (leaf) we fit a model
- ▶ This is typically a **constant** model, i.e. the average
- ▶ Why?
 - ▶ The piecewise constant model fit can be arbitrarily good (number of splits scales with data volume)
 - ▶ Making non-constant models consistent is computationally costly
- ▶ Why not?
 - ▶ Computational cost grows with tree depth
 - ▶ Often locally the structure is linear
- ▶ Leads to Local Regression Trees ¹
 - ▶ Complex if we ensure that the local regressions meet up

¹Karalic A, "Employing Linear Regression in Regression Tree Leaves" (1992)
ECAI-92

Discussion

- ▶ Could there be “discontinuities” where the model changes a lot?
- ▶ How could we prevent this?